

Physiological Targets of HIIT

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A well-known exercise training principle is that of progressive overload. The overload principle states that by working harder than is typically done in practice and creating stress, there will be a subsequent rebound adaptation in various physiological systems so that future stress is better tolerated (19). Thus, a general purpose of HIIT is to stress specific physiological systems used in sport to an equal or greater extent than that required during competition so that new levels of performance can be reached.

While a key focus of preparation for many sports must involve a high level of specific skill practice, the physical component of sport performance, such as that involved with completing a match, race, or competition, is often difficult to directly replicate day in and day out without running into major issues in preparation. As we alluded to in the opening chapter, replication of precise event stress at or above that which is occurring in an event or match is analogous to the mass destruction weapon approach, which in addition to being less specific to the adaptations required, also can run us toward problems that fall under the umbrella of overtraining, resulting in fatigue, injury, or some of the health aspects that will be emphasized in chapter 7. Thus, we need to go deeper into the science to help us solve our training program puzzle.

High-intensity interval training (HIIT) and its effects are complex. In general the scientific basis of HIIT is to produce a high-intensity overload stimulus

that offers the best signal to enhance the physiological capacity of certain systems of importance for the performance target in question. The next logical step therefore is to break down the HIIT response into distinct physiological categories that can become targets for our programming. The purpose of this chapter, then, is to understand the general physiological targets of importance for exercise performance, alongside the varying acute responses and potential adaptations to various forms of HIIT.

General Physiological Responses to HIIT

It is commonly accepted that many of the biochemical and physiological adaptations that accompany training occur in response to an increase in target cell energy demands (43, 73). Any exercise training session will challenge, at different respective levels relative to the training content, both the metabolic and the neuromuscular/musculoskeletal systems (34, 163). The metabolic system refers to three distinct yet closely related integrated processes, including

1. the splitting of the stored phosphagens (ATP and PCr),
2. the nonaerobic breakdown of carbohydrate (anaerobic glycolytic energy production), and

3. the combustion of carbohydrates and fats in the presence of oxygen (oxidative metabolism, or aerobic system) (64).

It is therefore possible to precisely characterize the acute physiological responses of any HIIT session based on the respective contribution of these three metabolic processes and the neuromuscular load and musculoskeletal strain.

As will be outlined in detail in chapters 4 and 5, the intensity, duration, number, and series of work and rest intervals that form a given HIIT session can be altered to tax particular metabolic pathways within muscle cells, as well as neuromuscular features. HIIT provides the signal to enhance oxygen delivery to target cells (muscle and nerve cells) for aerobic respiration in addition to its effect on altering the oxidative and glycolytic function of muscle cells. A more effectual locomotor profile then arises from training adaptations at both the cellular and systemic levels.

In this chapter, we will outline the general physiological responses that occur with HIIT. For simplicity and practicality, we have reduced the number of systems of focus to three of key importance and made specific reference to the aerobic oxidative system, the anaerobic glycolytic system, and the neuromuscular system (figure 3.1). As shown in figure 3.1, the anatomy and physiology of the human body involves a number of key metabolic systems and structures of importance. Any of these areas

may be of prominence for specific aspects of sport performance in question; however, the degree of involvement of these may depend on the associated sport's demands, the sport's weekly training pattern, or the individual athlete (chapter 1, figure 1.5). These important structures and systems, their function, and potential for adaptation will be elaborated on in this chapter.

Aerobic Oxidative System

Fundamentally, life, including sport performance, requires energy. Chemically speaking, this comes from the production and breakdown, or phosphorylation and rephosphorylation, of the energy molecule adenosine triphosphate (ATP). *Aerobic respiration* is the term we give to the generation of ATP via oxidative phosphorylation. As the name implies, oxidative phosphorylation requires large quantities of oxygen to nourish the cells of the body. Efficiency of this oxygen delivery is of particular importance to achieve peak-level exercise performance in most sports.

The cardiopulmonary system, made up of the lungs, heart, blood, and vessels, is the system that facilitates uptake of oxygen from the environment and its transport to the working cells of the body, primarily skeletal muscle, but additionally cardiac muscle and neural tissue, including the brain. Understanding the system and additionally methods of maximizing its function has been a focused research area for exercise physiologists over the last century.

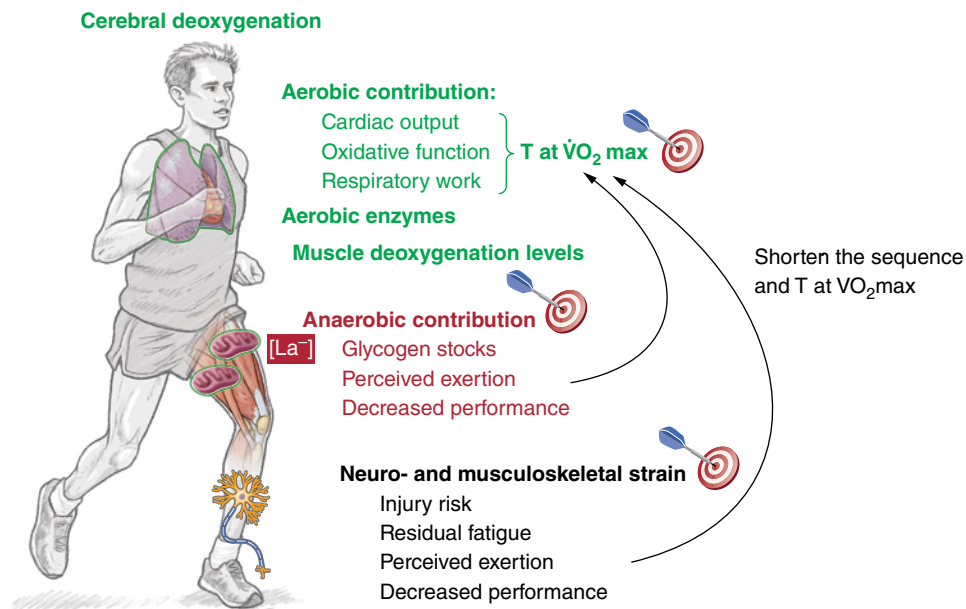


Figure 3.1 Key structural and metabolic targets of importance with HIIT, including the aerobic oxidative system, the anaerobic glycolytic system, and the neuromuscular system.

Integration of Maximal Cardiopulmonary and Cellular Oxidative Function: Time at $\dot{V}O_{2\max}$

At first blush, the perspective taken when one views the aerobic energy system is that it is the system responsible for life activities at rest and during steady-state submaximal activities. While that is true, what is less appreciated may be its importance at high metabolic rates. At the upper end of the spectrum, elite endurance athletes, such as runners, cyclists, and cross-country skiers, report maximal oxygen uptake ($\dot{V}O_{2\max}$) values that range from 70 to 85 mL·kg⁻¹·min⁻¹ (96). Achieving high rates of aerobic function, in both slow- and fast-twitch motor units, is critical for achieving high rates of energy production without fatigue. Thus, locomotor performance associated with $\dot{V}O_{2\max}$, and the duration of training time amounted at $\dot{V}O_{2\max}$, may be important. Indeed, $\dot{V}O_{2\max}$ might represent the integration of maximal cardiovascular and muscle metabolic (oxidative) function (26, 96).

HIIT protocols that elicit $\dot{V}O_{2\max}$, or at least a very high percentage of $\dot{V}O_{2\max}$, may provide the most effective stimulus for enhancing $\dot{V}O_{2\max}$. While evidence to justify the need to exercise at such an intensity remains unclear, it can be argued that only exercise intensities near $\dot{V}O_{2\max}$ allow for both the large motor unit recruitment (i.e., type II muscle fibers) (2, 68) and attainment of near-maximal ventilatory rates (89) and cardiac output, which, in turn, jointly signal for oxidative muscle fiber adaptation (including respiratory muscle) and myocardium enlargement (113) (and hence, $\dot{V}O_{2\max}$). For an optimal stimulus (and forthcoming cardiovascular and peripheral adaptations), it is believed that athletes should spend at least several minutes per HIIT session in their red zone, which generally means attaining an intensity of greater than 90% of $\dot{V}O_{2\max}$ (12, 103, 118, 119). Consequently, despite our limited understanding of the dose-response relationship between the training load and training-induced changes in physical capacities and performance (which generally shows large interindividual responses (21, 162)), there has been a growing interest from the sport science community to characterize training protocols that allow athletes to maintain the longest time above 90% $\dot{V}O_{2\max}$ (T at $\dot{V}O_{2\max}$, see Midgley and McNaughton (117) for review).

Manipulating parameters to maximize T at $\dot{V}O_{2\max}$ is complicated, and a more comprehensive analysis of the $\dot{V}O_{2\max}$ responses to different forms of

HIIT, from long intervals to SIT sessions through to short intervals and repeated-sprint sequences, will be the focus of chapter 5. Generally, both short (less than 60 s effort above v/p $\dot{V}O_{2\max}$) and long intervals (greater than 60 s effort at about v/p $\dot{V}O_{2\max}$) described in that chapter are effective methods of accumulating effective quantities (more than 10 min) of time at 90% to 95% $\dot{V}O_{2\max}$. While T at $\dot{V}O_{2\max}$ may be an important marker to observe when attempting to maximize performance potential, other physiological factors in isolation should also be examined.

Respiratory Function During HIIT

In addition to T at $\dot{V}O_{2\max}$, other targets in the cardiopulmonary system should also be considered to fully characterize the training stimulus when programming HIIT. The lungs as an organ are generally considered to be overbuilt structures, and in the healthy condition will typically not pose any real limitation to oxygen delivery (49). However, it has been reported that up to 50% of elite endurance athletes display a desaturation of arterial oxygen saturation at high exercise intensities near $\dot{V}O_{2\max}$, a phenomenon known as exercise-induced arterial hypoxemia (EIH) (133). The mechanisms describing EIH are variable and include a number of relatively unalterable factors including a ventilation-perfusion mismatch, transit time diffusion limitation, pulmonary edema, and a relative hypoventilation (49). In fact, there is some evidence that EIH develops as a natural consequence of gains in $\dot{V}O_{2\max}$ and endurance running ability following HIIT (122). Unfortunately, this occurrence, which may be due to a number of genetically derived factors creating individual differences in anatomy and physiology, is relatively unalterable. However, supplementation with blood buffers, such as sodium bicarbonate, may improve arterial saturation due to the Bohr effect and a right shift in the oxyhemoglobin desaturation curve (124). Furthermore, the occurrence of EIH in an athlete might also highlight an opportunity to attempt a legal blood volume enhancement strategy, such as using heat or altitude training (chapter 4). Thus, oxyhemoglobin saturation at $\dot{V}O_{2\max}$, alongside a hematological profile, can be measured in the elite endurance athlete to assess the potential for performance enhancement due to respiratory function limitations.

Cardiac Function During HIIT

A variable that is substantially more malleable, and the key facilitator of aerobic function, is the heart's

cardiac output, or the volume of blood that the heart can pump over a given time period. Cardiac output is typically expressed in liters per minute and has been reported to be as high as 40 L/min in elite athletes (53). As most sporting events are prolonged and require repeated high-intensity movement, it's not surprising that this key oxygen delivery system is considered by most as a fundamental variable of importance related to sport performance (96).

As maximal heart rate tends to be more or less fixed in athletes, not varying by more than 1 or 2 beats per minute each year (109), the main variable that adapts and changes with training tends to be the heart's stroke volume, or the volume of blood pumped with each beat (96). Stroke volume can be improved via two means. The first is through an increase in left-ventricular contractile force, while the other is through an increase in cardiac filling pressure (113), which increases end diastolic volume and resultant stroke volume via Starling's law (142). Improved cardiac filling pressure occurs with exercise, but also when the blood or plasma volume of the cardiovascular system is increased, as is the case following either exercise training or heat exposure, in which dehydration causes the kidney to retain salt and water in the cardiovascular system in the presence of raised serum aldosterone levels (42). However, the key signal

for making the chronic adaptations necessary, and the one we will focus on here, is the heightened cardiac filling pressure attained with HIIT.

Since reaching and maintaining an elevated cardiac filling pressure is believed to be necessary for improving maximal cardiac function (41, 46, 79), training at the intensity associated with maximal stroke volume may be warranted (104). Defining the key intensity accompanying maximal stroke volume is difficult, however, since this requires the continuous monitoring of the variable during exercise (70, 104, 106, 121, 135). Whether stroke volume is maximal at $v\dot{p}\dot{V}O_2\text{max}$, or prior to its occurrence, is still debated (see figure 3.2) (44, 69, 166). Stroke volume behavior during exercise is likely protocol dependent (115) and affected by training status (166), although this is not always the case (44). Moreover, there is only limited data available on cardiac function during exercise that resembles field-based HIIT sessions (70, 106, 121, 135). Studies have shown that maximal stroke volume is reached within ≈ 1 min (see figure 3.3) (121), ≈ 2 min (56, 70, 106), and ≈ 4 min (135), before decreasing (56, 70, 121) or remaining stable (106, 135) prior to fatigue. Nevertheless, inconsistencies with exercise intensity, individual training background, and particular hemodynamic behaviors (e.g., presence of a heart

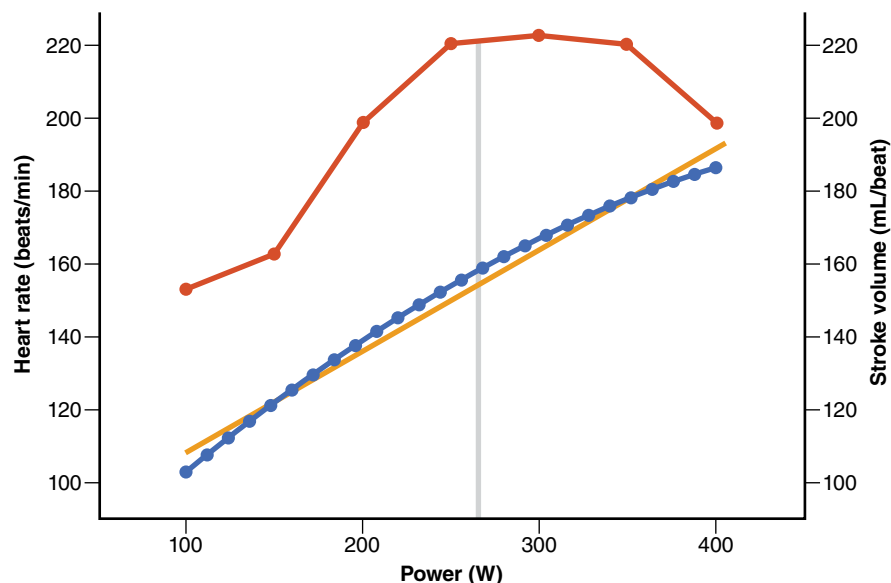


Figure 3.2 Time course of heart rate (HR; blue) and stroke volume (SV; red) during an incremental test in a subject presenting a typical break point in the HR-power curve. The orange line represents the HR responses if the relationship between HR and power output was linear.

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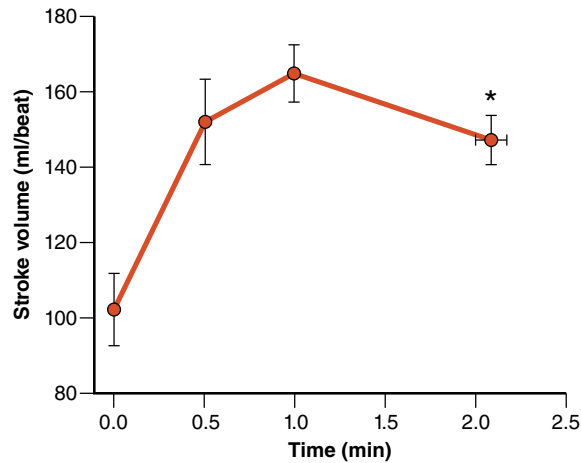


Figure 3.3 Stroke volume changes during constant load supramaximal cycling to exhaustion. Data are means $\pm$ S.E.M. for 8 subjects.

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rate deflection at high intensity (105)), as well as methodological considerations involved with measuring stroke volume, may contribute to these reported differences (44, 69, 166).

Based on the aforementioned studies measuring the time course of maximal stroke volume, it would appear, at least initially, that HIIT protocols involving long intervals (i.e., efforts greater than 60 s) should be applied in training when the goal is to enhance an athlete's maximal cardiac output (stroke volume). Based on this logic, short-interval HIIT (i.e., repeated 15 to 60 s bouts) would be assumed initially as a poor strategy unlikely to be effective for reaching maximal stroke volume and associated cardiac filling pressure, the short time course unable to trigger maximal stroke volume rates. Additionally, the alternating work and rest periods inherent with short-interval HIIT have been reported to induce variations in the action of the venous muscle pump, which may, in turn, limit the maintenance of a high stroke volume during the recovery period between intervals (84). Thus, logic to this point tells us that short-interval HIIT provides us with limited value if the goal of HIIT is to enhance maximal cardiac output. However, there are other aspects of HIIT sessions that we need to explore to confirm short-interval effectiveness, or lack thereof.

Particularly, we need to examine what happens to cardiac function during the rest or recovery periods

between HIIT bouts. In the 1970s, Cumming reported in fact that maximal stroke volume values were reached during the exercise *recovery* phase, and not during the exercise period itself, irrespective of the exercise intensity (45). While these results were measured during supine exercise in untrained patients, and despite contradictory claims (4), they contributed to the widespread belief that the repeated recovery periods and their associated high stroke volume accounted for the effectiveness of HIIT for improving cardiocirculatory function (59). In support of this, Takahashi et al. (157) found that stroke volume was 10% higher during the first 80 s of an active recovery (20% $\dot{V}O_{2\max}$) following a preceding submaximal cycling bout (60% $\dot{V}O_{2\max}$) in untrained males. Surprisingly, however, these hypothetical changes in stroke volume during recovery had never before been examined during typical HIIT sessions in athletes, and limited data is available examining the recovery intensity between HIIT work bouts.

To address some of these limitations, Buchheit, Racinais, and Girard (reported in Buchheit and Laursen (27)) measured hemodynamics during short-interval HIIT in a well-trained cyclist. Interestingly, their findings (figure 3.4) partly confirmed Cumming's results. Irrespective of the exercise chosen, i.e., incremental test, 3 min at $p\dot{V}O_{2\max}$, or repeated 15 s sprints (30% of the anaerobic power reserve, APR), stroke volume in a well-trained cyclist (PPO = 450 W, $\dot{V}O_{2\max}$ = 69 mL $\cdot$ min $^{-1}\cdot$ kg $^{-1}$) showed its highest values consistently during the recovery periods in his natural upright position on the bike. While acknowledging the limitations inherent to the impedance method used (PhysioFlow, Manaetec, France (38, 136)), these data suggest that despite its supramaximal nature, HIIT sessions may elicit cardiovascular adaptations via activity occurring specifically during the recovery periods.

Another uncertainty challenging the programming thoughts of practitioners and scientists surrounds the recovery intensity between HIIT bouts and what is best to maximize stroke volume. Should recovery be active and, if so, how active? Should it be passive? To examine this, Stanley and Buchheit (154) again used the PhysioFlow device to measure the effect of the recovery interval intensity on maximal stroke volume during 3 $\times$ 3 min work periods at 90% $p\dot{V}O_{2\max}$ followed by 2 min recovery intervals at either 0%, 30%, or 60% $p\dot{V}O_{2\max}$ in well-trained cyclists. Interestingly, stroke volume was equal across the three recovery intensities, including a passive recovery trial (figure 3.5). This important

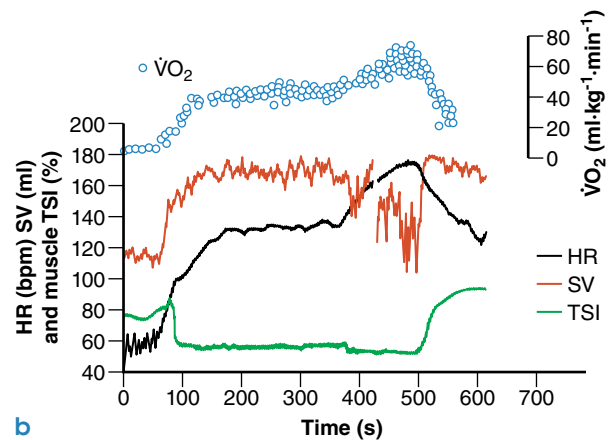
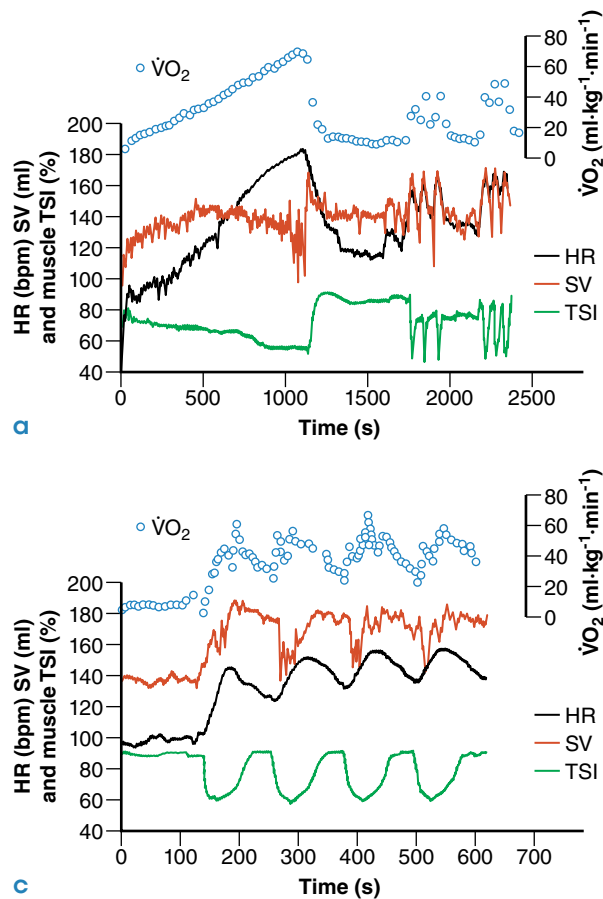


Figure 3.4 Oxygen uptake ($\dot{V}O_2$), heart rate (HR), stroke volume (SV), and muscle oxygenation (tissue saturation index, TSI) during an incremental test followed by (a) 2 sets of 3 supramaximal 15 s sprints (35% anaerobic power reserve, APR); (b) a 5 min bout at 50% of the power associated with $\dot{V}O_{2\text{max}}$ ($p\dot{V}O_{2\text{max}}$) immediately followed by 3 min at $p\dot{V}O_{2\text{max}}$; and (c) the early phase of an HIIT session (i.e., first four exercise bouts (15 s at 35% APR/45 s passive) in a well-trained cyclist. Note the reductions in SV for intensities above 50% of $\dot{V}O_{2\text{max}}$ during both the incremental and constant power tests. In contrast, maximal SV values are consistently observed during the post-exercise periods, either following incremental, maximal, or supramaximal exercises.

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finding tells us that exercise intensity during the recovery bouts separating HIIT should be relatively easy or passive if the purpose of the session is to complete additional high-intensity intervals and prolong accumulation of exercise at maximal stroke volume (with its associated high filling pressure).

The key takeaway messages from these data are that stroke volume can actually increase during the recovery bouts between HIIT work efforts, irrespective of the HIIT format (long or short), and the recovery intensity (degree of active versus passive recovery exercise) does not appear to add to the degree of maximal stroke volume attainment. As a result, recovery should be near passive to allow for work output during the HIIT effort phases to be maximized. To illustrate the effectiveness of short-interval HIIT for maximal stroke volume engagement, take for example an HIIT session involving 3 sets of 8×15 s sprint repetitions (30% of anaerobic power reserve, APR) interspaced with 45 s of passive recovery (long enough for peak stroke volume to be reached). Such a format would in theory allow an athlete to maintain his peak stroke volume for 24×20 s = 480 s, which is similar to the effort that

can be sustained during a constant-power exercise to exhaustion (106).

As a final note on the cardiovascular aspect of training, many authors, including yours truly (PL; (103)), have attempted to individualize and optimize between-work bout recovery duration using the return of HR to either a fixed value or percentage of its HR_{max} (1, 149). The present understanding of the determinants of HR recovery suggests, however, that this practice is not very relevant (145). During recovery, HR is neither related to systemic O_2 demand nor muscular energy turnover (32, 168) but rather to the magnitude of the central command and metaboreflex stimulations (142). That is, its response is complex and unlikely to be relevant within the context of achieving heightened levels of performance on subsequent HIIT work bouts. While it may be an important marker to monitor (chapters 8 and 9), its

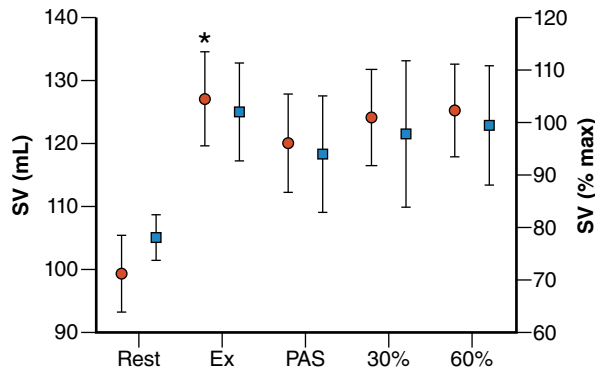


Figure 3.5 Stroke volume (SV) during different intensities during the HIIT session: preexercise rest (Rest), 3 min work periods at 90% $\dot{V}O_{2\max}$ (Ex), passive recovery (PAS), low-intensity recovery (30%), and moderate-intensity recovery (60%). Data are presented as mean ($n=14$; red circles) and percentage of incremental test peak values ($n=5$; blue squares). * indicates a small difference versus PAS. For clarity, the statistical symbols have been omitted for differences between rest and all other intensities as all differences are clear and large. Data, as a percentage of incremental test peaks, was provided to indicate the relative load and demonstrated similar trends to raw values.

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use for optimizing recovery during an HIIT session is currently not relevant.

In summary, reaching and maintaining an elevated cardiac filling pressure via maximizing time at high stroke volume may be a key stimulus for improving maximal cardiac function, and this appears to occur during both long and short intervals. As between-bout recovery intensity appears to have little to no effect on cardiovascular responses, athletes should be encouraged to reduce recovery intensity, toward very easy work or passive rest, when the aim is to increase work intensity or work duration at high intensity through a greater number of HIIT repetitions. Based on the information presented to this point, prescribing a variety of HIIT types to gain the stroke volume adaptation advantages of each exercise format might be best practice. HIIT sessions including near-to-maximal long intervals with long recovery durations (e.g., 3 to 4 min intervals and greater than 2 min recovery) might allow athletes to reach a high stroke volume during the work and

relief intervals. Even longer 4 min intervals at about 90% to 95% $\dot{V}O_{2\max}$ are also very popular (78, 79, 90). Alternatively, repeated short supramaximal work intervals (e.g., 15 to 30 s) with long recovery periods (greater than 45 s) might also be effective for reaching high values both during exercise (58) and in recovery. However, whether maximal cardiac output adaptations also occur following long- and short-interval sessions (i.e., continuous versus intermittent), or whether achieving a certain quantity of time at maximal cardiac output (T at $Q_{c\max}$) is needed to maximize its adaptation, is still unknown. In the only longitudinal study we are aware of comparing the effect of short versus long HIIT on maximal cardiovascular function (79), there was a *small* trend for greater improvement in maximal cardiac output for the long-interval protocol ($ES=+1$ versus $+0.7$ for 4×4 min versus $15/15$, respectively). Nevertheless, when the aim of the HIIT session is to maximize cardiac function, data to date suggests that prescribing a variety of short- and long-HIIT formats may fall under the category of best practice in most sports where this physiological component is important.

Muscle and Cerebral Oxygenation During HIIT

The repeated fluctuation of O_2 concentration in working muscle and brain during exercise training is likely to be one of a number of critical signals responsible for the improved muscular oxidative capacities and performance improvement shown following training (47, 48, 87). Such low levels of oxygenation during training are known to promote an improved energy state in working muscle, as indicated by a more-protected high-energy phosphate potential. By way of illustration, when Kilding and colleagues supplemented athletes with oxygen, so that they had likely improved tissue oxygenation during an HIIT session, it resulted in a *lowered* effect on improvements in physiological variables and subsequent performance compared with normoxic HIIT (97). To be clear, the participants improved less with the supplemental oxygen treatment during HIIT, as it likely lowered the local deoxygenation on the tissues of importance (muscle and nerve tissue), resulting in a lowered signal or stimulus for aerobic cellular adaptations (see below).

Strenuous and prolonged exercise additionally challenges brain oxygenation, both in normoxic (139) and hypoxic (160) conditions. Brain oxygenation is decreased at exhaustion at sea level during maximal (139) and supramaximal (120% to 150% $\dot{V}O_{2\max}$

(147, 148)) work. Low levels of brain oxygenation can trigger central fatigue in hypoxic environments (120), and may (152) or may not (15) be associated with impaired high-intensity exercise performance (160). Thus, manipulation of muscle and/or cerebral oxygenation to low levels are likely important features to consider when programming HIIT. To be clear, ensuring athletes are familiar with low levels of cerebral deoxygenation during training might help in delaying central fatigue and exhaustion during competition. Put another way, athletes must be comfortable with being uncomfortable. Whether training is achieved in a normoxic or hypoxic (i.e., at altitude) environment is typically not obligatory, unless of course competition occurs in such an environment, which is an entirely different topic. What does appear to be important, however, is that athletes achieve a certain degree of cerebral deoxygenation challenge during training. (See also environmental hypoxia section in chapter 4.)

Muscle oxygenation changes during exercise are clearly affected by different HIIT variables. Muscle oxygenation, as measured by near-infrared spectroscopy (NIRS), reflects the balance between O_2 demand and delivery. Therefore, the greater the mismatch between O_2 demand and delivery, the greater the deoxygenation (74, 75). Whether muscle $\dot{V}O_2$ during exercise is more related to metabolic inertia or reduced muscle blood flow and O_2 delivery is still debated (36, 72, 95). It is evident, however, that muscle deoxygenation levels increase in parallel with increases in exercise intensity during incremental testing (9), although direct comparisons between exercises of different intensities are limited (6, 125, 146). The rate of deoxygenation is likely exercise intensity dependent, with higher exercise intensity eliciting faster deoxygenation levels (6), but contrasting results have also been reported (146). One clear effect of supramaximal (all-out) work performed during HIIT, repeated-sprint sequences (RSS), and sprint interval training (SIT) sessions is that it enables large levels of muscle (23, 25, 33) and cerebral (147, 148) desaturation to be reached. For example, the absolute muscle deoxygenation reported during a SIT session (including 30 s sprints) is profound (Δ tissue oxygenation index, $TOI = -27\%$), and similar to values observed during interventions known to produce near-to-maximal muscle desaturation (i.e., maximal voluntary leg contractions (37) or post 2 min upper-body limb occlusion (98)). Indeed, muscle deoxygenation during SIT (~30 s all out) may be two-fold greater than during shorter

(i.e., 4 s,) repeated maximal running sprints ($\Delta TOI = -12\%$ (23)).

For a given supramaximal work intensity (i.e., 120% of $v/\dot{p}\dot{V}O_{2max}$), the longer the work interval, the greater the deoxygenation (39, 40, 158). However, close inspection of the data presented by Turner et al. (figure 3.6 (158)) suggests that, with a work:relief ratio of 1:2 (and a 20 W relief intensity), the magnitude of muscle deoxygenation does not increase further after 30 s of exercise (i.e., Δ deoxyhemoglobin $\approx 6, 28, 32$, and $32 \mu M$ for 10, 30, 60, and 90 s efforts at 120% $\dot{p}\dot{V}O_{2max}$). In the studies by Christmass and colleagues, higher deoxygenation levels were reached after 40 s, compared with 12 (39) or 26 s (40) of exercise, while a trend toward a plateauing in deoxyhemoglobin was evident during the latter portion of the 40 s analysis (40). Taken together, these results suggest that to reach maximal muscle (and probably cerebral) deoxygenation levels, all-out efforts lasting 30 to 40 s may be preferred. A long between-sprint recovery (i.e., 4 min), as generally implemented during typical SIT sessions (35, 88), is probably not required to repeatedly reach these levels of deoxygenation. When such efforts are interspersed with 2 min of passive rest (25, 31), a complete reoxygenation can occur and the magnitude of (at least muscle) deoxygenation has been shown to increase with sprint repetitions (30), likely related to greater energy requirements from fatigued muscles per unit of external work (8, 172). Finally, if we consider that repeated fluctuations of muscular deoxygenation levels during training sessions are likely key signals for improving muscular oxidative capacities (47, 48), and with greater exercise intensities eliciting the greatest deoxygenation levels, it seems prudent to recommend periods of high-amplitude HIIT to achieve the necessary training adaptations (i.e., HIIT including near-to-maximal effort intensity interspersed with passive rest).

In summary, the key HIIT targets to maximize oxygen delivery and uptake often involve strategies that prolong T at $\dot{V}O_{2max}$, lengthen time under high cardiac filling pressure and maximal stroke volume, as well as strategies that cause large fluctuations in muscle and/or cerebral oxygenation levels. In chapter 5, we will further explore how different HIIT parameters can be manipulated to emphasize these specific physiological HIIT target types.

Metabolic Systems in Skeletal Muscle

Like all cells in the body, muscle cells require energy to do their job. Sport performances often exemplify

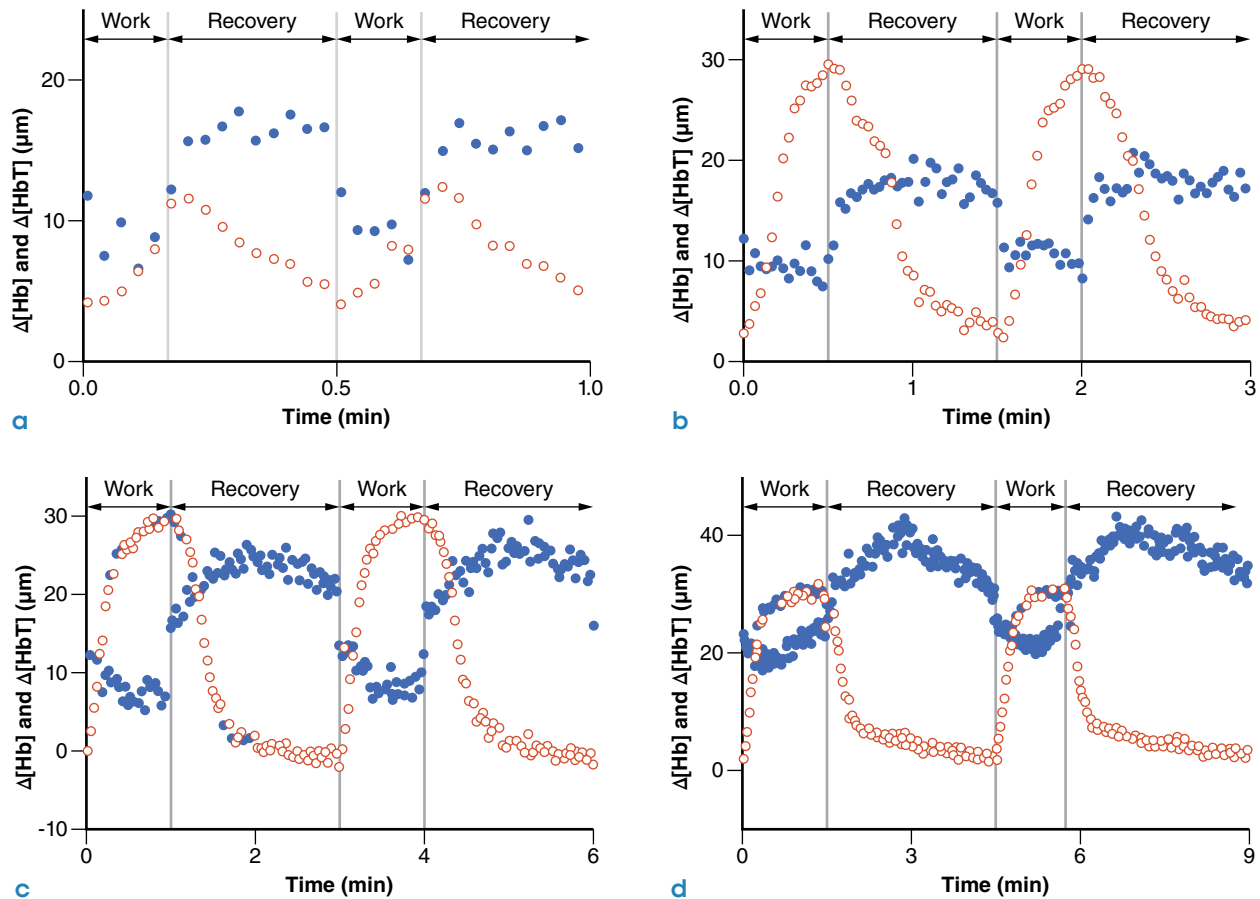


Figure 3.6 Representative responses in a single subject of deoxygenated hemoglobin (open circles) and total hemoglobin (filled circles) in the vastus lateralis muscle. Changes in hemoglobin and deoxygenated hemoglobin expressed as change from baseline to four intermittent exercises: (a) 10 s effort/20 s of rest; (b) 30 s/60 s; (c) 60 s/120 s; (d) 90 s/180 s. The exercise intensity was set at 120% of $\dot{V}O_2$ max for the four intermittent exercises.

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some of the highest rates of muscular metabolic energy transfer in modern life. The efficiency of energy production in skeletal muscle may be determined by an integration of the metabolic pathways supplying ATP (capacity of glycolytic and oxidative enzymes) and the excitation-contraction processes that utilize it (73). Following various forms of HIIT, both the oxidative enzyme pathways responsible for aerobic respiration, as well as the anaerobic glycolytic enzyme system important for generating ATP at rapid rates without oxygen, are typically upregulated (55, 76, 80, 107, 110, 130, 137, 167).

A well-functioning aerobic oxidative system, referring to the energy pathways involved with producing ATP aerobically in muscle, is often of great importance for sport performance. We just learned of the importance of oxygen delivery to cells via the

cardiopulmonary system for high-intensity exercise performance. Increased capillarization of those engaged muscle cells, a process termed *angiogenesis*, which increases the surface area of engagement between the cardiovascular system and the muscle cell, is also needed to facilitate efficient oxygen extraction (85). However, we have been more or less referring to these delivery systems in isolation, when in reality there must be a receiving end to that oxygen delivery, with fuel, predominantly glucose and fatty acids, able to be burned in those muscle cells in the presence of oxygen to generate energy for movement. The majority of this burning, or oxidation of fuel, happens specifically in the cell's mitochondria, via a number of metabolic pathways and enzyme reactions. Interestingly, current consensus holds that the functional condition and quantity of an organism's

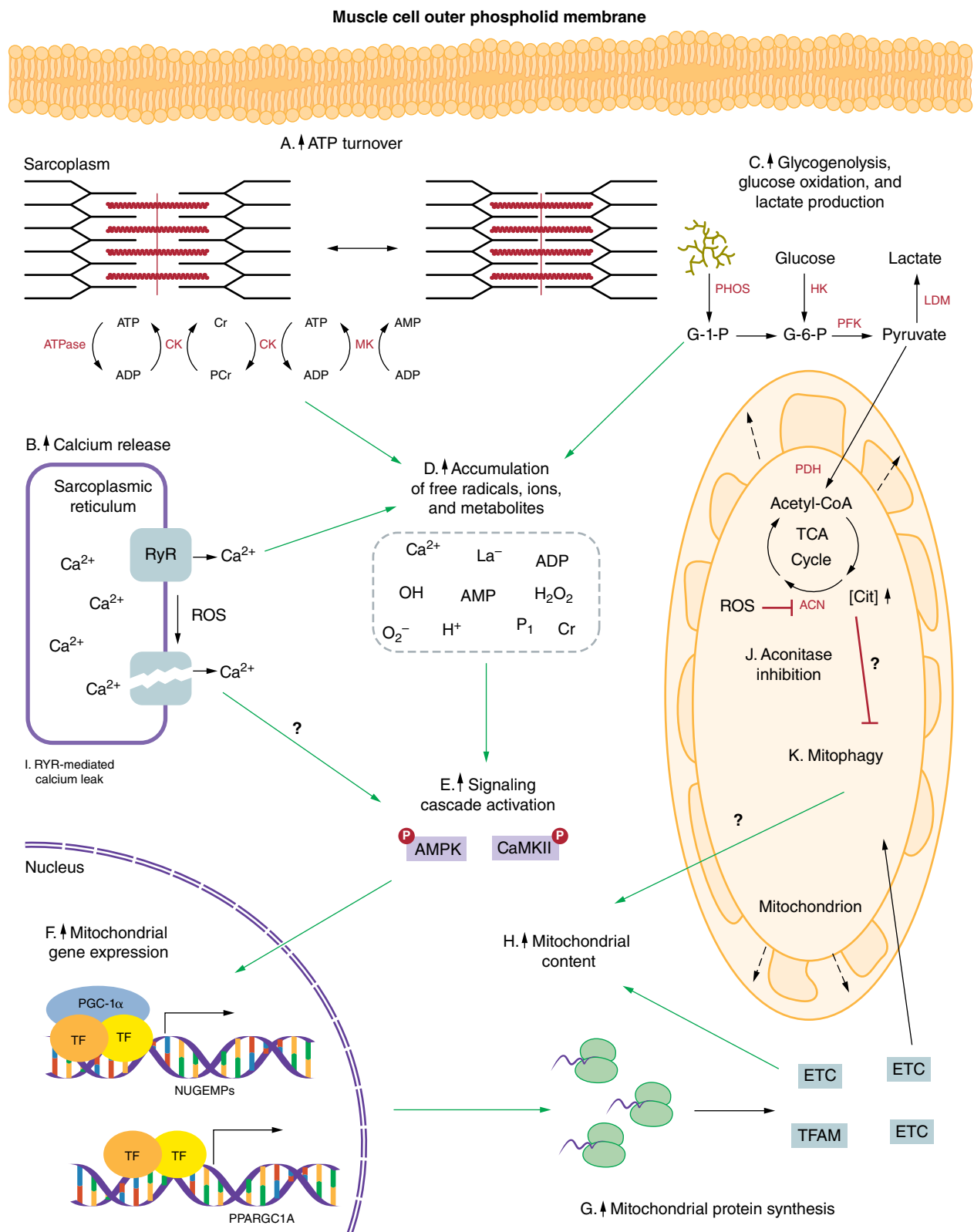


Figure 3.7 The mechanisms through which high-intensity exercise may elicit greater mitochondrial adaptations to aerobic training compared to lower intensities of exercise. Exercising at a higher intensity requires greater ATP turnover (A), increases calcium release (B), and involves greater use of carbohydrates for fuel (C), compared to exercising at a lower intensity. As a result, there is a greater accumulation of metabolites, ions, and free radicals (D), which increase the activation of signaling proteins (E), including Ca²⁺/calmodulin-dependent protein kinase II (CaMKII) and AMP-activated protein kinase (AMPK). The increased activity of these protein kinases causes greater rates of gene

mitochondria, the energy powerhouses of our body's cells, determines the health and longevity of a person (108), and may be one of many reasons to ensure that sport and/or exercise is part of a daily routine throughout the lifespan.

Figure 3.7, from MacInnis and Gibala (111), provides a schematic representation of a muscle cell and its key working parts impacted by HIIT, and provides an important foundation for the physiological HIIT targets that will be described throughout the book.

While figure 3.7 provides the microlevel perspective for the inner workings of a body's muscle cell when it receives the HIIT stimulus, figure 3.8 gives us a more macro whole-body perspective, and for this example, we present data from an elite Australian male road cyclist. During submaximal continuous exercise, there is a relatively stable and constant ATP turnover in the cell that is met by the oxygen source we've just described. Under this steady rate condition (i.e., 65% $\dot{V}O_2$ max), there is a balanced level of carbohydrate and fat oxidation used to meet the ATP turnover needed. When higher energy demand is called on, as it is with high-intensity exercise, there are broadly two new conditions that can occur in the muscle based on the degree of intensity required by the nervous system. First, as intensity increases with continuous exercise, a point occurs above which the higher ATP turnover rate is usually met by a greater degree of carbohydrate over fat oxidation. This intensity is typically called the *aerobic threshold* and may represent the maximal rate of the body's fat oxidation. The first lactate threshold, or 1 mmol/L increase in blood lactate above baseline levels, or first *ventilatory threshold* (VT_1 ; first increase in minute ventilation relative to oxygen uptake), is the progressive exercise test marker of this point (figure 3.8; (123)). Second, under the con-

dition at which energy demand is greater than the ATP turnover that can be met aerobically with oxygen (i.e., critical speed/power intensity or anaerobic threshold; see chapter 4), the ATP turnover can be temporarily met via an increase in glycolytic rate without oxygen (anaerobic), with pyruvate accepting the protons needed to allow the chemical process to continue generating energy. Subsequently, lactate and hydrogen, or lactic acid, is formed and diffused into the bloodstream (figure 3.8). This maximal lactate steady state, or critical power, can also be seen as the point at which there is a substantial increase in the minute ventilation relative to the volume of carbon dioxide production (bicarbonate buffering of lactic acid).

Recall that above our critical power, VT_2 or maximal lactate steady state, is the exercise intensity that constitutes an HIIT session. Such high-intensity exercise induces a number of downstream actions in muscle cells as a consequence to the action that may be important to mention due to the adaptation signals they may induce. When a muscle's motor neuron initiates action to contract, the cell's sarcoplasmic reticulum increases calcium release in muscle to signal for the contraction process to occur, which causes actin-myosin filaments to move across one another and shorten (figure 3.7). When the energy demand across larger and larger motor units is initiated, as it is with various HIIT forms, and for longer durations, this demands a greater use of carbohydrates to be used as fuel compared to lower-intensity exercise. Thus, acutely, high-intensity exercise above critical speed/power, such as HIIT, specifically draws on greater rates of carbohydrate oxidation, predominantly from stored glycogen reserves. The high carbohydrate use with HIIT is important to appreciate, as with finite reserves of glycogen, HIIT programming

Figure 3.7 (continued)

expression for PGC-1 α , which in turn acts as a transcriptional coactivator for nuclear genes encoding mitochondrial proteins (F). In turn, mitochondrial protein synthesis rates are greater for high-intensity exercise (G), leading to a greater increase in mitochondrial content (H) relative to exercise at a lower intensity. Two additional reactive oxygen species (ROS)-mediated mechanisms include fragmentation of the ryanodine receptor (RyR) of the sarcoplasmic reticulum and increased intracellular calcium concentrations (I), a signal for mitochondrial biogenesis. Inhibition of aconitase (J) in the tricarboxylic acid cycle (TCA) and an increase in intracellular citrate concentration has been suggested to increase mitochondrial content via a reduction in mitophagy with HIIT. ACN: aconitase; ATPase: adenosine triphosphatase; CK: creatine kinase; ETC: electron transport chain; PHOS: glycogen phosphorylase; HK: hexokinase; LDH: lactate dehydrogenase; MK: myosin kinase; PFK: phosphofructokinase; PDH: pyruvate dehydrogenase; TFAM: transcription factor A mitochondria.

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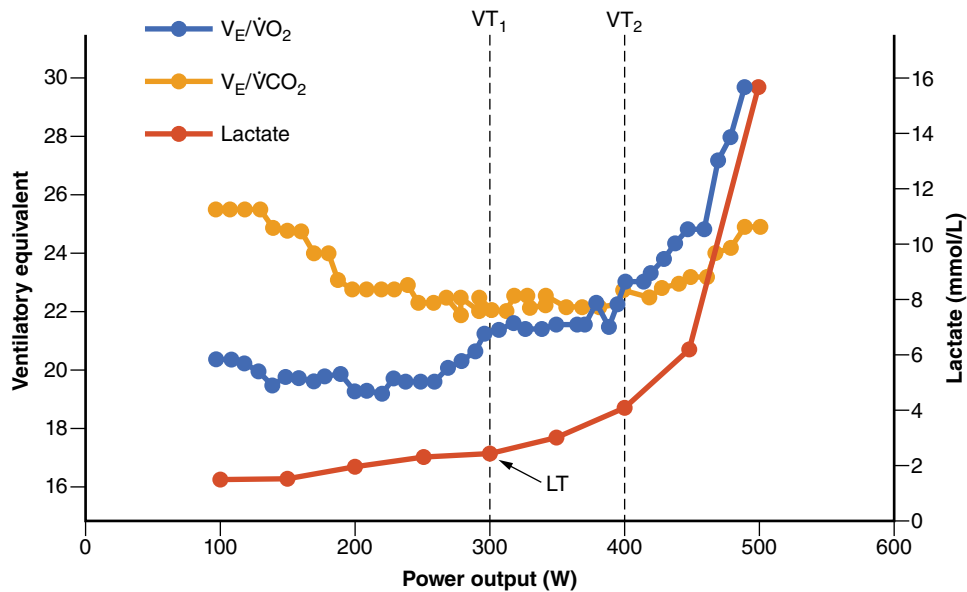


Figure 3.8 Various anaerobic threshold demarcation points during a progressive exercise test (100 W+50 W/5 min) in an elite Australian cyclist ($\dot{V}O_{2\max}=78.2 \text{ mL}\cdot\text{kg}^{-1}\cdot\text{min}^{-1}$). The first ventilatory threshold (VT_1) and lactate threshold (LT) occurred at 300 W, while the second ventilatory threshold (VT_2) and/or respiratory compensation point occurred at 400 W.

Adapted by permission from R.U. Newton, P.B. Laursen, and W. Young, "Clinical Exercise Testing and Assessment of Athletes," in *Olympic Textbook of Medicine in Sport*, edited by M.P. Swellnus (Oxford, UK: Blackwell Publishing, 2008), 160-199.

needs to be appropriately managed with this in mind so as to solve the puzzle appropriately (see sections in chapters 1, 4, and 5).

In addition to the larger draw on energy reserves during high-intensity exercise, which lowers the ratio of ATP:ADP, metabolites, ions, and free radicals also accumulate in muscle. Together these acute demands increase the activation of signaling proteins, with the two most noteworthy including the kinases CaMKII and AMPK (111). The increased activity of these protein kinases causes far greater rates of gene expression from the so-called "master switch," the PGC-1 α , which in turn acts as a transcriptional coactivator for nuclear genes encoding mitochondrial proteins (111). As a result, HIIT, in a variety of forms, has been shown to enhance mitochondrial protein synthesis rates, leading to a greater increase in mitochondrial content, fat oxidation rates, GLUT4 transporters, glycogen content, and oxidative fiber-type conversion (figure 3.9) (5, 102).

Indeed, the density of mitochondria in skeletal muscle may be the fundamental regulator of substrate metabolism during exercise, with increased mitochondrial content promoting a greater reliance on fat oxidation and a proportional decrease in carbohydrate oxidation (86). Following adaptation,

compared to an equivalent bout of exercise done prior, glycogen degradation and lactate production at a given intensity will be reduced, with lactate threshold increased, thus allowing individuals to exercise for longer durations and at greater percentages of their $\dot{V}O_{2\max}$ (96). Practically speaking, exercise feels easier at the same work rate, amounting to less fatigue and faster recovery (161). As we described in chapter 1, practitioners must be mindful of the anaerobic training volume in their programs, due to its potential to create fatigue aspects that might carry over and interfere with subsequent training content (i.e., skill-based work or other high-intensity sessions, and potentially contributing to overtraining, chapter 7). Thus, training that emphasizes aerobic energy turnover, with limited anaerobic engagement, tends to be prioritized for most practitioners and coaches, whose aim is to instill a good level of fatigue resistance in their athletes. Generally, sessions with a large aerobic energy involvement and little neuromuscular stress elicit low overall fatigue, thereby permitting athletes to execute their tactical and skill-based training content appropriately. Aerobic-based metabolic sessions, in which the byproducts of energy production are for the most part simply CO_2 and water, should be emphasized

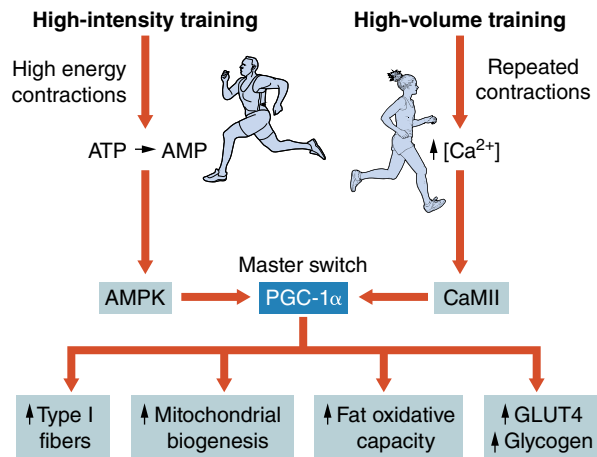


Figure 3.9 Simplified model of the adenosine monophosphate kinase (AMPK) and Ca^{2+} calmodulin-dependent protein kinase II signaling pathways, as well as their similar downstream target, the peroxisome proliferator-activated receptor- γ coactivator-1 α (PGC-1 α). This “master switch” is thought to be involved in promoting the development of the aerobic muscle phenotype. High-intensity training appears more likely to signal via the AMPK pathway, while high-volume training appears more likely to operate through the CaMK pathway. Both signals, from a blend of distinct periods of high- and low-intensity exercise, may be important. ATP: adenosine triphosphate; AMP: adenosine monophosphate; GLUT4: glucose transporter 4; Ca^{2+} : intramuscular calcium concentration.

Adapted by permission from P.B. Laursen, “Training for Intense Exercise Performance: High-Intensity or High-Volume Training?” *Scandinavian Journal of Medicine & Science in Sports* 20 Suppl 2 (2010): 1-10.

with a high level of implementation in nearly every sport’s program (chapter 5; type 1, 2, 3, 4 targets).

Anaerobic Glycolytic System

While the cardiorespiratory and cellular aerobic oxidative systems, and to a lesser extent muscle deoxygenation, tend to be the main variables of interest for understanding HIIT format design, anaerobic glycolytic energy contribution (see (14, 34, 163)) is another important parameter to consider. Increases in ATP turnover rate via heightened rates of glycolytic capacity may facilitate important improvements in short-duration sport performance. Glycolytic enzyme activity, in particular that of lactate dehydrogenase, phosphofructokinase, and glycogen phosphorylase, has been shown to increase after both

short (less than 10 s) and long (greater than 10 s) duration sprint interval training (140), alongside improved physical work capacity (107, 110, 130, 137). While the plasticity or ability to change the capacity of the anaerobic energy system with HIIT appears less compared to the aerobic oxidative system, the anaerobic energy contribution to high-intensity exercise may change more due to the enhanced ability of the neuromuscular system to engage a greater number of larger motor units, including more fast-twitch muscle fibers (discussed subsequently).

While a noninvasive method of assessing anaerobic glycolytic energy contribution to high-intensity exercise has not been definitively established (116), the maximal accumulated oxygen deficit (MAOD) method (64) and muscle lactate concentration rate estimate (99) tend to be the preferred methods. The MAOD method, however, has a number of limitations (7, 12, 126) and such data have only been reported in a few HIIT-related studies (156, 163). Methods used to estimate MAOD are variable, but best practice is currently thought to involve using 10×4 min submaximal exercise bouts to establish the power output: $\dot{V}\text{O}_2$ relationship before solving for the y-intercept value based on the performance power output achieved using a sport-specific supra-maximal exercise (figure 3.10) (126).

While the MAOD method has been referred to where possible, blood lactate accumulation can also be reported as a surrogate marker of anaerobic glycolytic capacity (28). The use of blood lactate concentration to assess anaerobic glycolytic energy contribution has a number of limitations, including large individual responses, prior nutritional substrate status (171), session timing in relation to prior exercise (92), timing of sampling post exercise (112), the effect of aerobic fitness (11), and its poor association with muscle lactate (93), especially following high-intensity intermittent exercise (99). Nevertheless, since all subjects would generally be expected to present with normal nutritional/substrate stores when involved in a study, the potential influence of these latter factors for the studies reviewed in our book is likely to be low. Therefore, with the aforementioned limitations in mind, we have used blood lactate changes during exercise to estimate anaerobic energy contribution for a given exercise (10). In an attempt to compare the anaerobic glycolytic energy contribution during different forms of HIIT, we will focus herein on post-HIIT values and on the initial rate of blood lactate accumulation in the first 5 min of exercise (134). This latter measure was selected

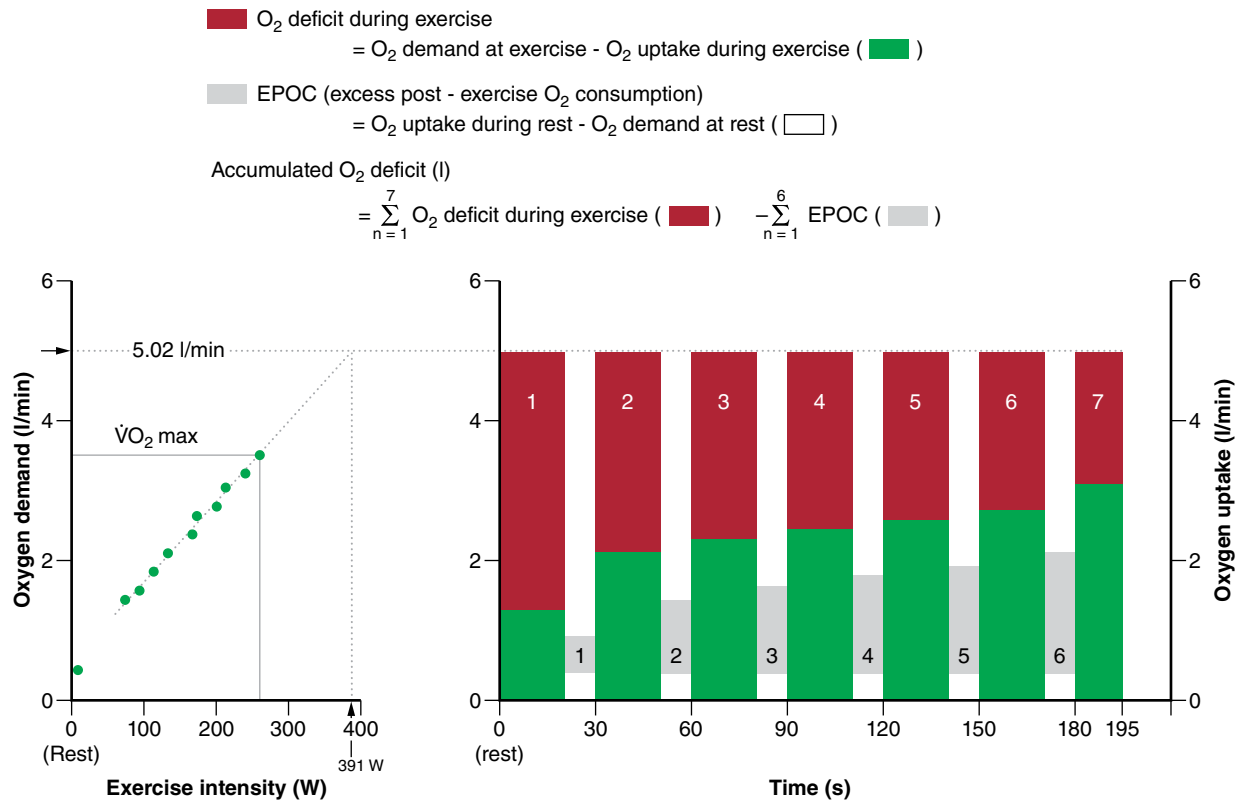


Figure 3.10 Maximal accumulated oxygen deficit (MAOD) method of anaerobic glycolytic energy measurement estimation. For the intermittent exercise, the oxygen demand was taken from the linear extrapolations of the oxygen demand versus power output relationship established during a pretest. The oxygen deficit during each bout of the exercise was taken as the difference between the oxygen demand and the oxygen uptake during the intermittent exercise. The excess post-exercise oxygen consumption (EPOC), which reflects a recovery of the body's oxygen stores and possibly some resynthesis of phosphocreatine during the rest periods, was calculated as the difference between measured oxygen uptake during the rest periods and the resting oxygen demand. The oxygen demand at rest between each bout of exercise was set equal to the resting oxygen uptake measured before the experiment. MAOD during exhaustive intermittent exercise was calculated to be the difference between the sum of the oxygen deficit during each bout of the intermittent exercises and the sum of the EPOC during the rest period between bouts of intermittent exercise.

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due to the fact that blood lactate values collected after prolonged HIIT bouts do not permit discrimination between different HIIT sessions. For example, at exhaustion in any exercise bout, participants have already reached a plateau in their blood lactate accumulation (51, 52). The 5 min duration was chosen since it corresponds to the average time-to-exhaustion shown for continuous exercise at $\dot{V}O_{2\max}$ (13, 82), is near the duration of long-interval bouts used with typical HIIT sessions, as well as being the duration of most RSS (chapters 4 and 5). Additionally, this duration approximates the time needed for blood

lactate levels to normalize as a function of the metabolic demand (10, 77). For reviewed studies in which blood lactate values were not provided ~5 min following exercise onset, the rate of blood lactate accumulation was linearly extrapolated to a predicted 5 min value using pre- and post- (immediate) exercise measures. (Only exercises lasting 2 to 6 min were included in this analysis.) Post-HIIT blood lactate values were categorized as very low (less than 3 mmol/L), low (3 to 6 mmol/L), moderate (greater than 6 mmol/L to 10 mmol/L), high (greater than 10 mmol/L to 14 mmol/L), and very high (greater

than 14 mmol/L). HIIT sessions were also categorized based on the initial rate of blood lactate accumulation as follows:

- strongly aerobic:** less than 3 mmol·L⁻¹·min⁻¹;
- aerobic:** greater than 3 mmol·L⁻¹·min⁻¹;
- mildly anaerobic:** greater than 4 mmol·L⁻¹·min⁻¹;
- anaerobic:** greater than 5 mmol·L⁻¹·min⁻¹;
- strongly anaerobic:** greater than 6 mmol·L⁻¹·min⁻¹ (28).

Controlling the level of anaerobic glycolytic energy contribution during HIIT sessions is another important programming consideration we can use to optimize an athlete's preparation. There are likely two main scenarios to bear in mind when it comes to selecting the appropriate anaerobic contribution of an HIIT sequence: limiting or enhancing the anaerobic contribution, for various purposes. In the first scenario, such as during busy training schedules, when there is little time to recover between sessions (certain distance runners training twice a day, subsequently performing large volumes of HIIT, or in team sport players juggling field work, gym sessions, and matches), coaches may prefer a lower volume of lactic sessions to avoid overload, allowing players to be ready to perform during frequent competitions (100). In contrast, at some key moments during the team sport season, and for a number of sports in which a high glycolytic energy contribution and/or the ability to tolerate high levels of acidosis are required (e.g., track-and-field sprint athletes, some team sports), coaches may actually prefer to program these lactic sessions. As described by Iaia and Bangsbo (88), so-called *lactic training* is designed to stimulate the anaerobic glycolytic system and involves periods of exercise performed at intensities above $v\dot{p}V\dot{O}_2\text{max}$. Lactic training may be further divided into a number of different formats, broadly defined as pure speed training, speed endurance training, or speed endurance maintenance training (88) (chapter 5; type 3, 4, 5 targets). Finally, the programming of HIIT with a high anaerobic contribution can also be viewed from the broader training perspective, where it's not the session response per se that matters most but rather its effect on glycogen stores and the subsequent training sequences that may occur. In fact, training low (169, 170) (i.e., when glycogen stores are low) is becoming more popular and has been shown to enhance aerobic adaptations in the muscle (mitochondrial biogenesis; figures 3.7 and 3.9) (94). While there are

many ways to induce glycogen depletion to allow athletes to train low, such as dietary manipulation and fasting (see chapters 4 and 7), using endurance training sessions (169) or more importantly, HIIT with high energy contributions, may be the more controllable options. The nuances of glycolytic energy contribution and adaptation following HIIT are complex, as a result of the various parameters that can be manipulated with HIIT, and these distinctions will be expanded on in chapter 4.

Neuromuscular System

Every movement made by athletes requires communication between the brain and the muscles. The neuromuscular system, which includes the muscles of the body and the nerves serving them, enables this coordination. The fundamental unit of the nervous system, the motor unit, includes the motor nerve and all of the fibers it innervates. The ratio of motor nerve to fiber number explains our fine versus gross motor movement (54).

When we refer to the musculoskeletal and neuromuscular load associated with HIIT, we mean the various physical stressors an athlete's anatomy encounters during a training session and the acute effects this has on both the neuromuscular and musculoskeletal systems. Such stresses could include, among others, the tension developed in locomotor muscles, tendons, joints, and bone; the muscle fiber recruitment and the associated changes in neuromuscular performance that may occur as a function of the potential neural adjustments; and changes in force-generating capacity that occur during training. Indeed, the high exercise intensities (i.e., $\geq 90\%$ to $95\% v\dot{p}V\dot{O}_2\text{max}$ to 100% of maximal sprinting speed or power), the changes of directions for some types of HIIT performed in racket and team sports, and the large volumes (runners can cover 6 to 8 km at $v\dot{V}\dot{O}_2\text{max}$ per session) of HIIT sessions lead naturally to high engagement of both neuromuscular and musculoskeletal systems. HIIT-induced neuromuscular fatigue is a critical component that can influence a multitude of other important training content, from the ability to complete in full or with high quality any number of training components that could be considered key in an athlete's preparation, including:

1. the HIIT session itself, which can subsequently reduce T at $\dot{V}\dot{O}_2\text{max}$ and, in turn, limit the ability to reach the desired metabolic adaptations (see chapter 5),

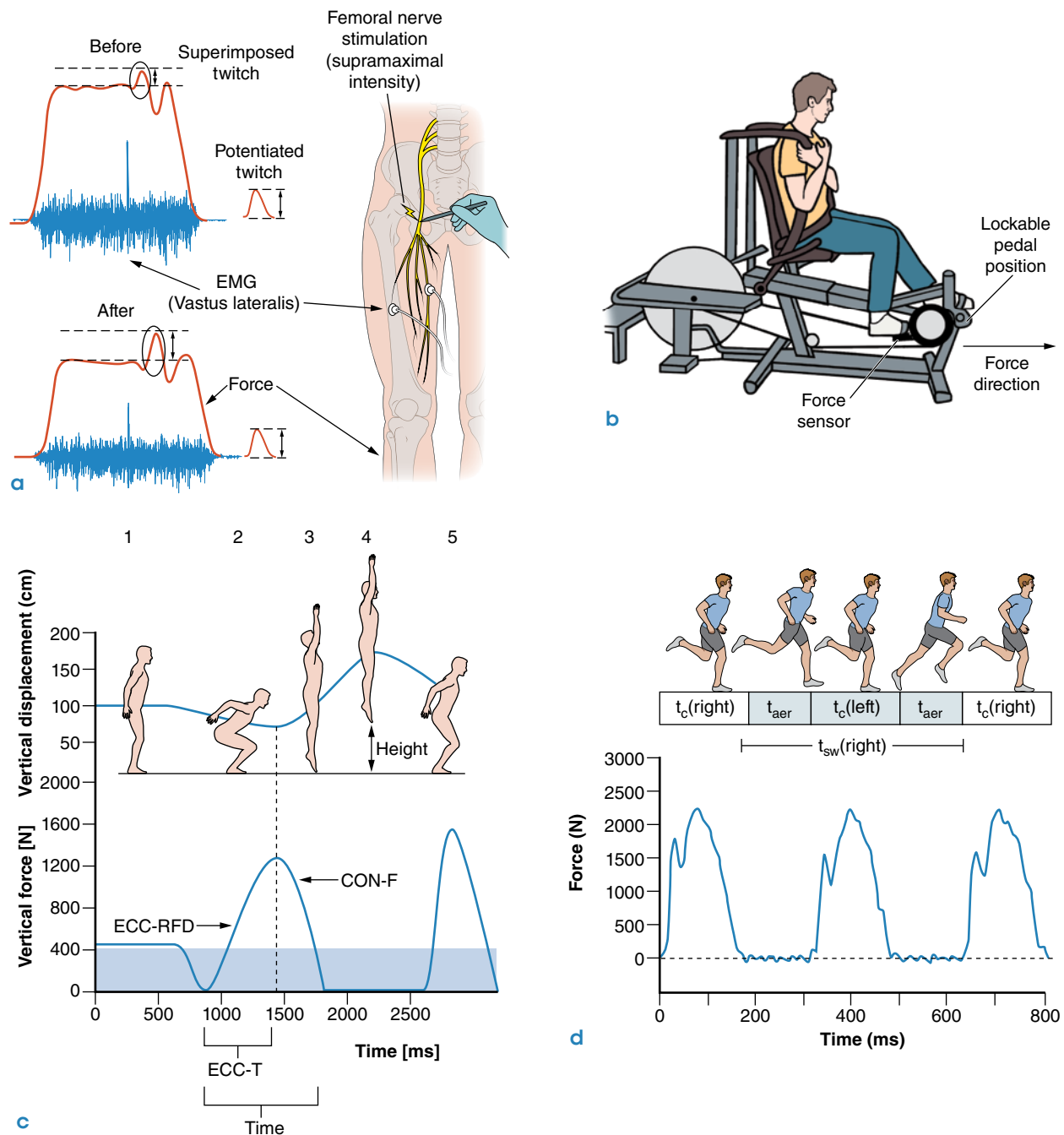


Figure 3.11 Tools used to measure the neuromuscular system response: (a) maximal voluntary contraction (MVC) on a classical leg-extension ergometer and the associated force and EMG responses, still considered the gold standard measurement of neuromuscular status, as additional neural stimulation allows the full characterization of muscle activation levels. This MVC was taken before and after a repeated-sprint cycling exercise (10×6 s with 30 s recovery). By comparing the twitch superimposed to an MVC and the twitch evoked on the relaxed muscle (i.e., femoral nerve supramaximal stimulation), the twitch interpolation technique in conjunction with surface EMG (i.e., root mean square [RMS] value normalized by the maximal muscle compound action potential [M-wave]) can noninvasively and reliably characterize muscle activation. Voluntary activation level (%) was estimated according to the following formula: $(1 - [\text{superimposed twitch}/\text{potentiated twitch}] \times 100)$. Note that both the voluntary activation level and the normalized RMS activity were depressed (−2.5% and −14.5%, respectively) compared with values obtained before exercise. (b) Innovative ergometer used to assess neuromuscular fatigue during cycling exercise (159). The pedal position is lockable and the transition from cycling to an isometric maximal voluntary contraction of the knee extensors at ~90 degree pedal position is lockable and the trans direction of applied force. (c) Countermovement jump (CMJ) and

2. a certain volume of low-intensity aerobic training due to HIIT-related locomotor fatigue (144), which would be expected to complement the overall training adaptations,
3. the execution of important skill training and associated learning (potential carryover effects from previous training sessions (19, 141)),
4. a resistance training session (chapter 6),
5. a carryover (20) or interference effect (60, 61), which can limit strength and explosive power development (chapter 6), and many other aspects related to injury risk, including traumatic-type injuries (127, 151) and overuse-type injuries both in runners (138) and team sport athletes (153).

The topic of interference is so important that we have an entire chapter devoted to it, written by a leader in the field, Dr. Jackson Fyfe (chapter 6), with specific reference to the complex integration of HIIT and resistance training.

In practice, coaches must choose between two main options: enhancing or limiting neuromuscular load during the HIIT sessions in their programs. At certain times they will seek to increase the neuromuscular characteristics of the HIIT sessions in an attempt to improve the athletes' locomotor function (i.e., running or movement economy) and, hypothetically, the fatigue resistance of the lower limbs (neuromuscular learning effect (20)). Many methods exist to stress the neuromuscular system specifically. Performing uphill HIIT sessions; climbing stairs in a stadium; running in the sand (12, 16, 62); introducing jumps, lunges, sit-ups; performing short shuttle runs or lateral running between work intervals (so-called Oregon circuits) (50); slowing cadence during cycling (131); adding eccentric cycling; and adding weights or bungees during rowing or kayak

training are all variations coaches have used to make gains and add stress to the neuromuscular system in varying sports. Nevertheless, neuromuscular strain, if maintained for several hours or days after the HIIT session, can have a direct effect on the quality of subsequent training sessions (both neuromuscular-oriented as strength or speed sessions), as well as lend to the possible interference phenomenon (60), or impact on technical and tactical sessions in team sports, not to mention the most important psychological component (see chapter 9).

In contrast to endurance athletes, team sport players generally tend to perform low-volume HIIT sessions with minimal acute neuromuscular strain or fatigue, so that athletes can gain freshness for upcoming tactical, strength, and speed sessions (24). In the final phase of team sport competition preparation, however, a high neuromuscular load during HIIT might also be needed in players to replicate specific game demands (81, 84). Finally, if we consider that a good athlete is first an injury-free healthy athlete, neuromuscular load also must be considered within the context of musculoskeletal pain and injury risk management. Running speed, time run at high intensity, as well as specific running patterns or ground surfaces, all known to influence the neuromuscular system, must also be considered when programming HIIT (28). HIIT type 2, 4, and 5 targets all involve neuromuscular engagement, and the refinements of the HIIT weapons to achieve these response types are elaborated on in chapter 5.

As we have just outlined, the impact that various forms of exercise training have on the neuromuscular system is important for a number of reasons. Despite the importance of controlling neuromuscular and musculoskeletal load during HIIT sessions, there has been limited data quantifying the effect of HIIT variable manipulation on neuromuscular

Figure 3.11 (continued)

associated force trace on a force plate that is used to calculate various indices such as fly time, peak power, velocity at takeoff, or rate of force development, all of which share various levels of association with neuromuscular fatigue (65). (d) Force trace during running, often used to assess stride parameters and leg stiffness, which is affected by neuromuscular fatigue.

Figure 3.11a: Reprinted by permission of Springer Nature from O. Girard, A. Mendez-Villanueva, and D. Bishop, "Repeated-Sprint Ability - Part I: Factors Contributing to Fatigue," *Sports Medicine* 41 no. 8 (2011): 673-694. Figure 3.11b: Adapted from R. Twomey, S.J. Aboodarda, R. Kruger, S.N. Culos-Reed, J. Temesi, and G.Y. Millet, "Neuromuscular Fatigue During Exercise: Methodological Considerations, Etiology and Potential Role in Chronic Fatigue," *Neurophysiologie Clinique/Clinical Neurophysiology* 47 no. 2 (2017): 95-110. (c) 2017 Elsevier Masson SAS. All rights reserved. Figure 3.11c: Adapted by permission from G. Laffaye, P.P. Wagner, and T.I.L. Tombleston, "Countermovement Jump Height: Gender and Sport-Specific Differences in the Force-Time Variables," *Journal of Strength and Conditioning Research* 28 no. 4 (2014): 1096-1105. Figure 3.11d: Reprinted by permission from P.G. Weyand, D.B. Sternlight, M.J. Bellizzi, and S. Wright, "Faster Top Running Speeds Are Achieved With Greater Ground Forces Not More Rapid Leg Movements," *Journal of Applied Physiology* 89 no. 5 (2000): 1991-1999.

function. Data on neural and muscular adjustments using force trace measures and motor nerve stimulation (101, 132, 150, 165) following high-intensity exercise suggests that fatigue induced by HIIT including either very short (less than 20 s) to short (≤ 1 min) or nonmaximal efforts ($\leq \sim 120\%$ $\dot{V}O_{2\max}$) tends to be predominantly peripheral in origin (i.e., alterations to excitation-contraction coupling) (101, 132). In contrast, muscle fatigue during repeated long (i.e., ≥ 30 s) and all-out sprints may be more related to central mechanisms (57). However, few authors have investigated the neuromuscular responses to HIIT sessions using sport- and training-specific tasks, with maximal isometric voluntary contractions of the active musculature (i.e., maximal voluntary contractions, MVC) generally chosen as the laboratory-based task for assessment of neuromuscular changes (101, 132, 150, 165). This is problematic since (muscle) fatigue is task-specific (54). To partially overcome this task-related limitation (155), some researchers have developed a new ergocycle (figure 3.11b) that allows the measurement of maximal isometric force production during cycling-specific movements. In fact, in addition to the typical cycling mode, the pedals can be locked instantly in a fixed position, comparable to the standard position used during isometric evaluation. With this setup, participants' cycling can be brought to a complete stop within 1 s in the designated position for isometric evaluation. Isometric force can then be measured during voluntary and evoked contractions by a previously validated wireless pedal-force analysis system located between the pedal and the crank. For run-based sports, alternate and perhaps more-specific field-based measurements might include the assessment of changes in counter-movement jump (CMJ) height and sprint speed (22, 128, 143), musculoskeletal stiffness regulation (50, 67), and stride parameters (66). While CMJ height reflects the efficiency of both muscle activation and muscle contractile properties (143), sprint speed can be a less-precise measurement of neuromuscular fatigue, as changes in inter- and intramuscular coordination factors, motor control, and/or stride parameters may limit the speed decrement during maximal sprints, so that fatigue appears to be less pronounced. For example, jump performance shows a much greater performance decrement (i.e., 3 times) than sprint performance during repeated sprint and jump sequences (29). This contrasts, however, with recent findings showing sprinting times and maximum theoretical horizontal velocity to be

more affected than CMJ performance following a fatiguing Rugby Sevens session (114). Differences may be related to the type of fatiguing exercise (repeated sprint and jump sequence versus full rugby session) and timing of the testing (during the sequence itself versus before and after a rugby session). Another limitation of these latter performance measures is that they are generally of maximal nature, while in practice HIIT with short and long intervals is often not performed at maximal intensity (i.e., less than maximal sprinting speed or peak power output; chapters 4 and 5). It is also worth noting that the acute effects of high-intensity running on leg muscle performance depend on the physiological characteristics and training history of the athlete (22, 71, 128). For example, while explosive athletes involved with speed events (i.e., track and field) or team sports generally show impairments in muscular performance following high-intensity exercises (63, 71, 128), endurance-trained athletes tend to show less impairment (71), no changes (63, 128), or even improvements (164) (possible postactivation potentiation (83)). During repeated-sprint exercises, similar findings have been reported, with endurance-trained athletes showing less fatigue than team sport athletes (18). Finally, with the exception of a very limited number of studies (17), post-exercise neuromuscular tests are generally performed either immediately after or 10 to 30 min following the HIIT sessions. Since residual fatigue is likely to be extended following such sessions, investigating the time course of muscular and neural responses to HIIT over a longer time course (i.e., hours/days (3, 91, 129)) is needed to assess potential carryover effects of HIIT on the subsequent training sessions (19). Figure 3.11 shows the tools used to measure the neuromuscular system response, including the MVC, cycling-MVC, CMJ, and stride parameters.

Summary

In summary, this chapter described the three key general physiological response targets involved with HIIT: the aerobic oxidative system, with particular reference to the delivery (cardiac output/stroke volume) and oxidative (mitochondrial) energy production components; the short-term anaerobic glycolytic energy system; and the various aspects affecting the neuromuscular and musculoskeletal systems. The next chapter details the intricacies of manipulating HIIT parameters and the ways these factors affect the acute physiological responses.